

AKM Sabbir

Interactive Project for Software Engineering CS616

Date: 02/28/2015

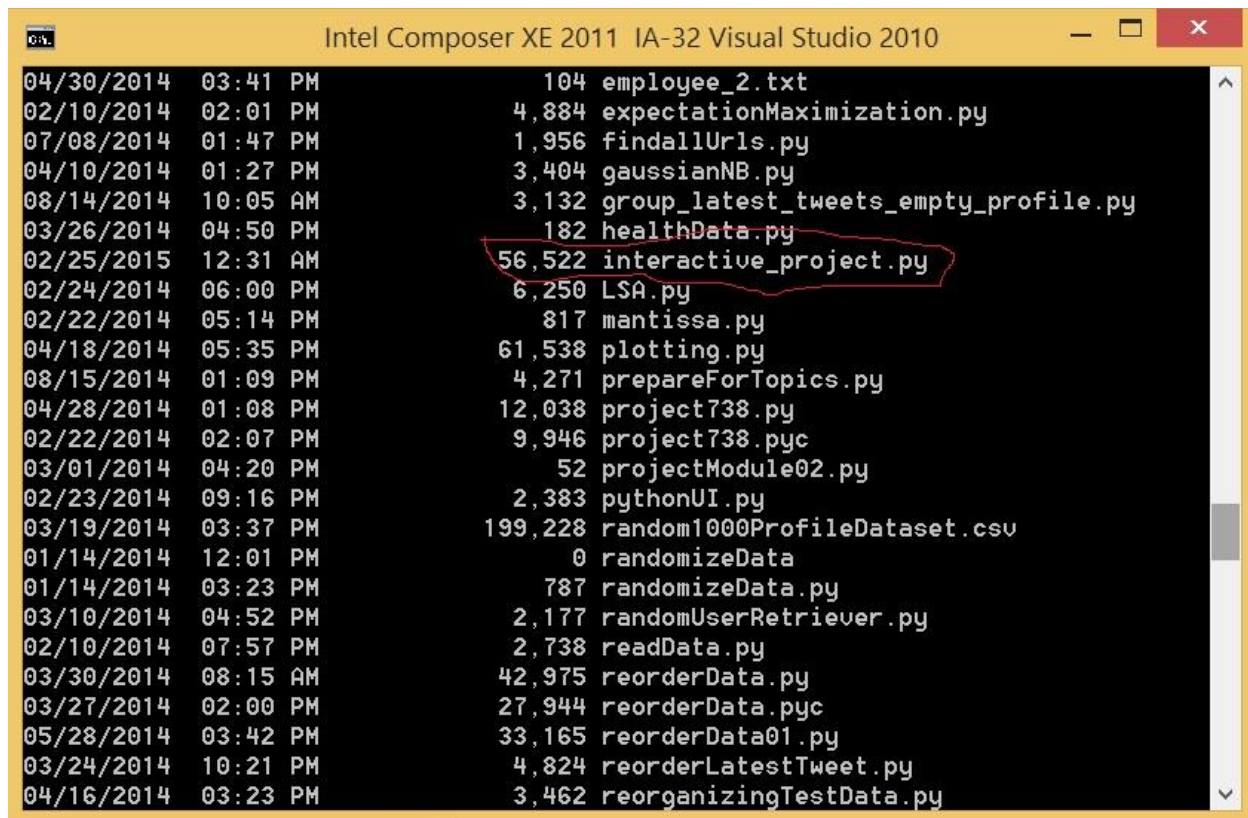
Tutorial Application:

This 'tutorial point' desktop application has been developed for the students of 6th – 10th grade so that they can learn about structural programming, programming language and IOI submission using languages like C++,python etc. It is developed by using python 2.7 and wxPython. It is platform independent so the program can be run in Linux, Windows, MacOS platform. By using this application students can prepare for competition like the International Olympiad in Informatics. It will help students to learn how to solve easy level programming problems using programming code and also help students to practice critical thinking. To run the application one has to perform the following steps first,

1. Download and install python 2.7 from <https://www.python.org/download/releases/2.7.5/>
2. Add installation directory to Environment Variable. If it is windows add the directory to Path variable and if it is Linux then export it to PATH variable.
3. Then download and install wxPython from <http://www.wxpython.org/download.php>
4. Open a command prompt go to the directory where application script exist and run the following command:

Python interactive_project.py

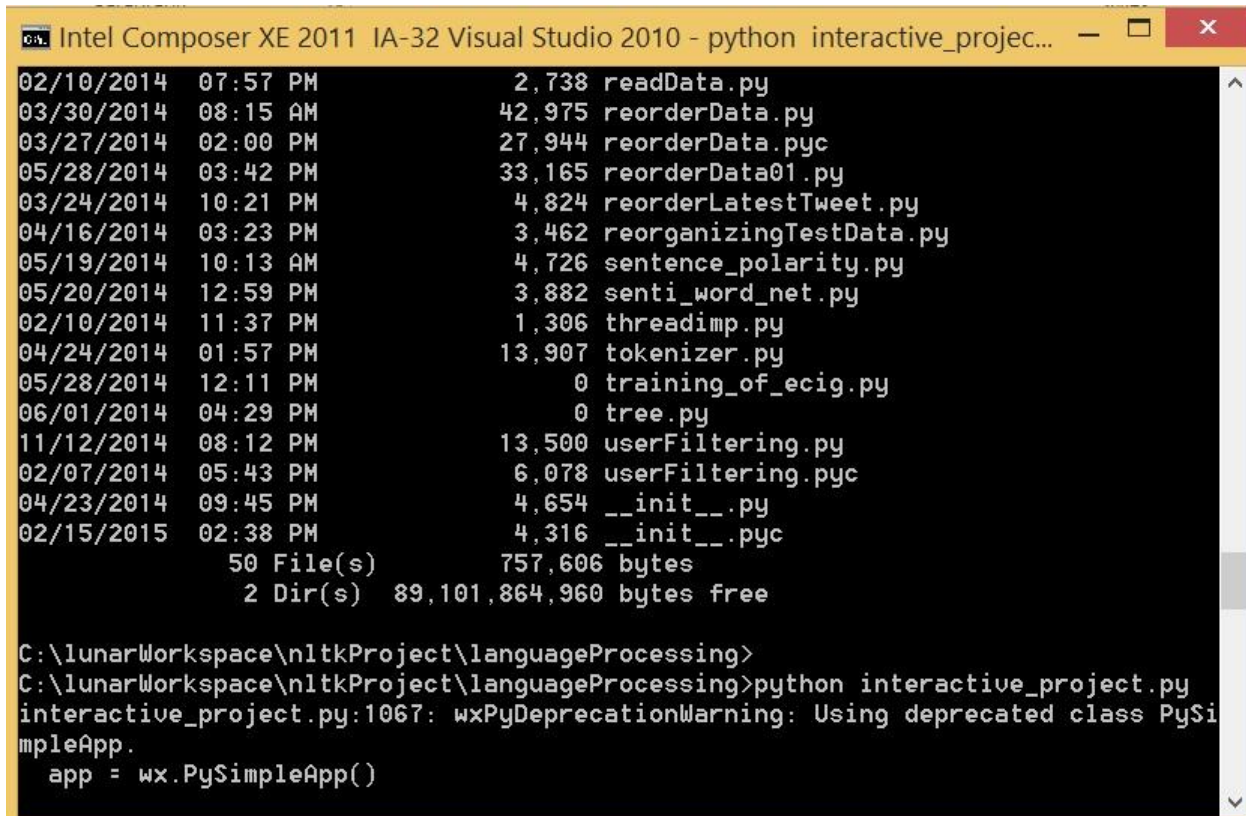
Following images is going to show details of it,



Date	Time	Size	File Name
04/30/2014	03:41 PM	104	employee_2.txt
02/10/2014	02:01 PM	4,884	expectationMaximization.py
07/08/2014	01:47 PM	1,956	findAllUrls.py
04/10/2014	01:27 PM	3,404	gaussianNB.py
08/14/2014	10:05 AM	3,132	group_latest_tweets_empty_profile.py
03/26/2014	04:50 PM	182	healthData.py
02/25/2015	12:31 AM	56,522	interactive_project.py
02/24/2014	06:00 PM	6,250	LSA.py
02/22/2014	05:14 PM	817	mantissa.py
04/18/2014	05:35 PM	61,538	plotting.py
08/15/2014	01:09 PM	4,271	prepareForTopics.py
04/28/2014	01:08 PM	12,038	project738.py
02/22/2014	02:07 PM	9,946	project738.pyc
03/01/2014	04:20 PM	52	projectModule02.py
02/23/2014	09:16 PM	2,383	pythonUI.py
03/19/2014	03:37 PM	199,228	random1000ProfileDataset.csv
01/14/2014	12:01 PM	0	randomizeData
01/14/2014	03:23 PM	787	randomizeData.py
03/10/2014	04:52 PM	2,177	randomUserRetriever.py
02/10/2014	07:57 PM	2,738	readData.py
03/30/2014	08:15 AM	42,975	reorderData.py
03/27/2014	02:00 PM	27,944	reorderData.pyc
05/28/2014	03:42 PM	33,165	reorderData01.py
03/24/2014	10:21 PM	4,824	reorderLatestTweet.py
04/16/2014	03:23 PM	3,462	reorganizingTestData.py

Fig 1:

As I run the dir command in the current directory it shows my project file interactive_project.py along with some other project file. now I am going to run python_interactive project.py

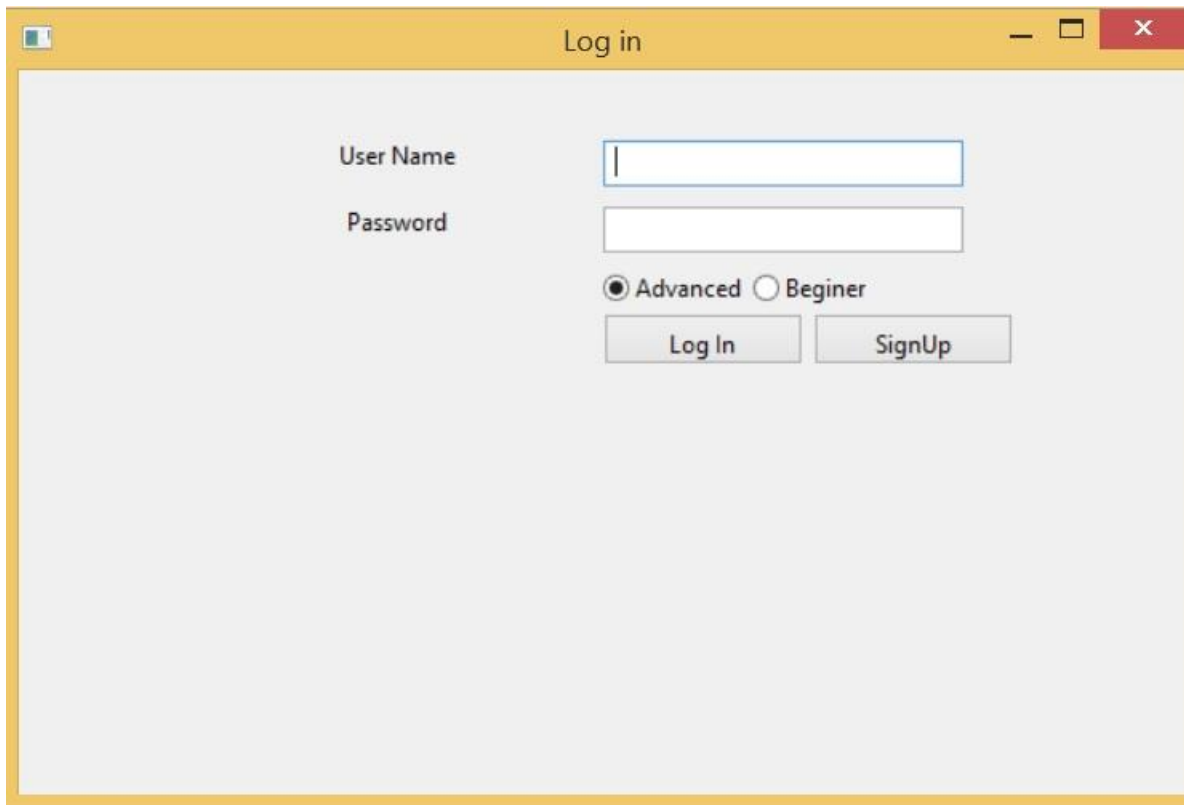


```
Intel Composer XE 2011 IA-32 Visual Studio 2010 - python interactive_projec...
02/10/2014 07:57 PM          2,738 readData.py
03/30/2014 08:15 AM        42,975 reorderData.py
03/27/2014 02:00 PM        27,944 reorderData.pyc
05/28/2014 03:42 PM        33,165 reorderData01.py
03/24/2014 10:21 PM         4,824 reorderLatestTweet.py
04/16/2014 03:23 PM         3,462 reorganizingTestData.py
05/19/2014 10:13 AM         4,726 sentence_polarity.py
05/20/2014 12:59 PM         3,882 senti_word_net.py
02/10/2014 11:37 PM         1,306 threadimp.py
04/24/2014 01:57 PM        13,907 tokenizer.py
05/28/2014 12:11 PM           0 training_of_ecig.py
06/01/2014 04:29 PM           0 tree.py
11/12/2014 08:12 PM        13,500 userFiltering.py
02/07/2014 05:43 PM         6,078 userFiltering.pyc
04/23/2014 09:45 PM         4,654 __init__.py
02/15/2015 02:38 PM         4,316 __init__.pyc
          50 File(s)          757,606 bytes
           2 Dir(s)  89,101,864,960 bytes free

C:\lunarWorkspace\nltkProject\languageProcessing>
C:\lunarWorkspace\nltkProject\languageProcessing>python interactive_project.py
interactive_project.py:1067: wxPyDeprecationWarning: Using deprecated class PySi
mpleApp.
  app = wx.PySimpleApp()
```

Fig 2:

As soon as user runs the command it starts the wx.PySimpleApp() and generate the following login GUI,



Log in

User Name

Password

☒ Advanced ☐ Beginer

Log In SignUp

Fig 3:

So any user who want to use the application has to sign up for an account and create the necessary username and password to log into the application. It has two options advanced level for upper grade students and beginner level for mid grad students. Right now I am using the following username and password and advanced level option,

Username: akmsabbir

Password: 0189

After login user will get the following main graphical user interface,

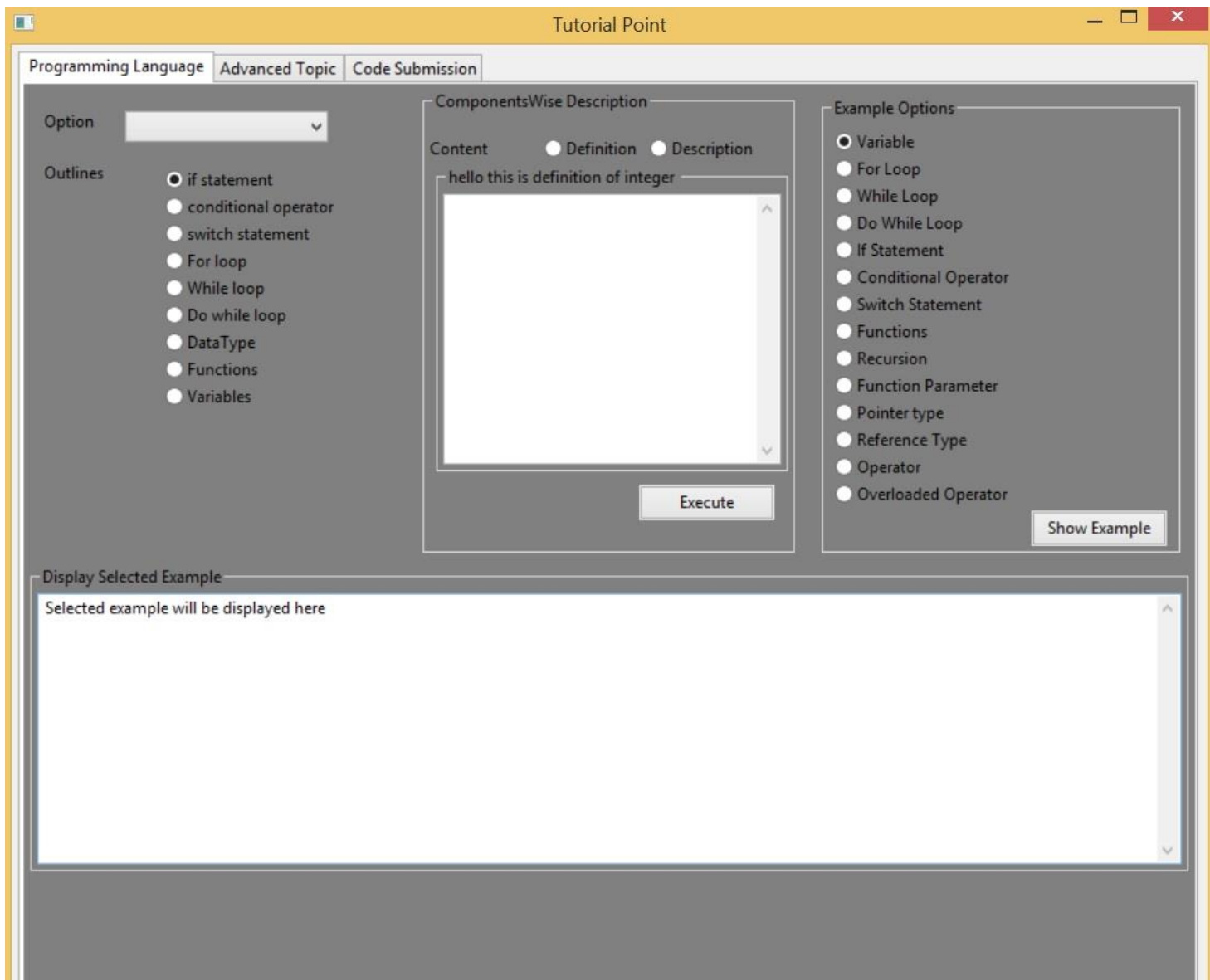


Fig 4:

So the main graphical user interface has three tabs. Each tab covers a specific domain of the problem of learning about programming, programming language, IOI. It might have more tabs covering more topics and area however this is a big project and in a short time I tried to make it concise and demo level presentation. So the different components and options available here are described below:

1. Option field helps the user to narrow down his search space for information he is looking for. The available options are ,
 - i. Loops
 - ii. Variable
 - iii. Datatype
 - iv. Conditional Statement

v. Functions

For instance if user chooses the option Loops he will get the following window,

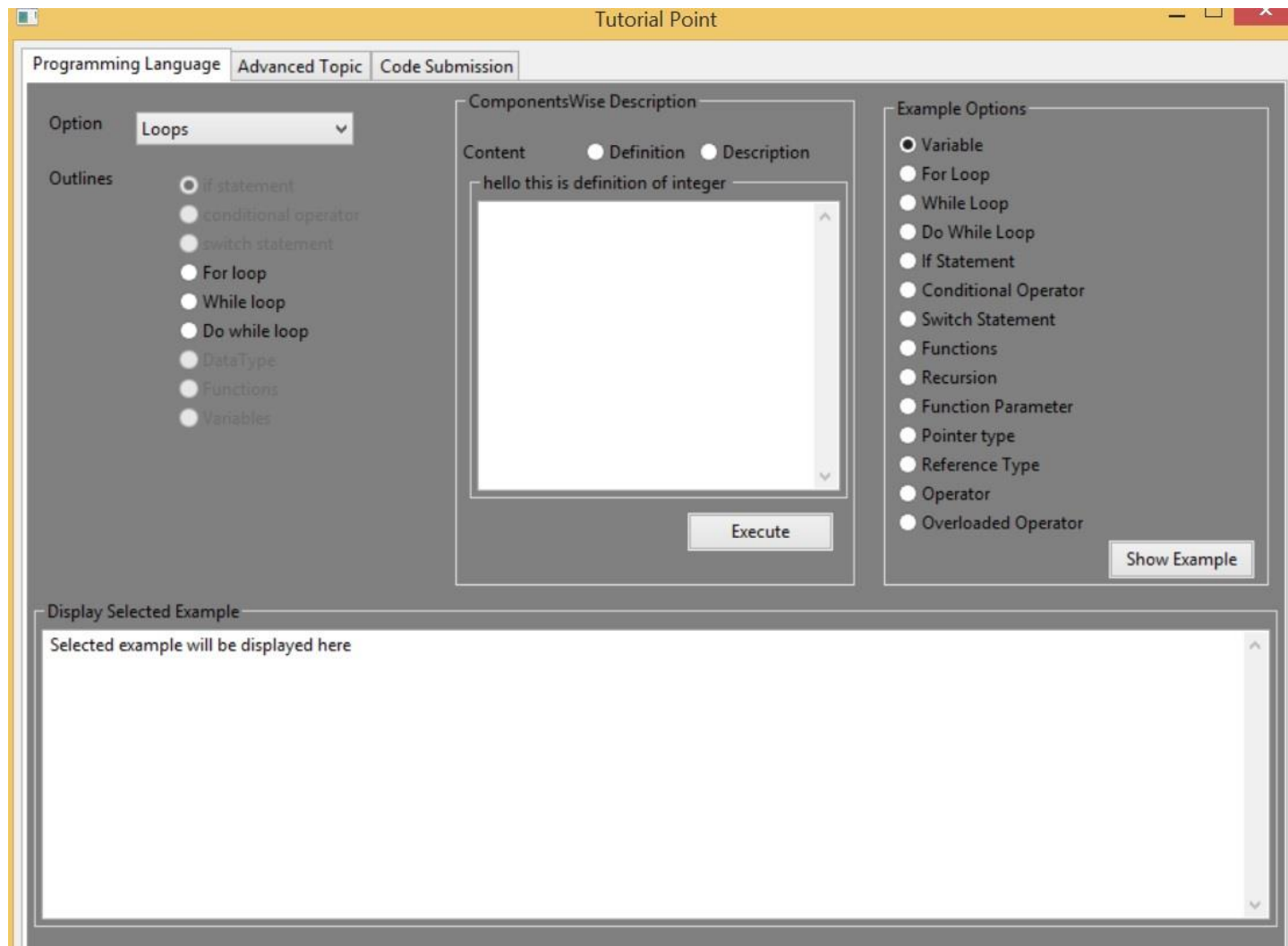


Fig 5:

So in the outline area user has three loop options to choose from. Now user can select either content Definition or Description and press the execute button to get more information about it. As the following image shows that,

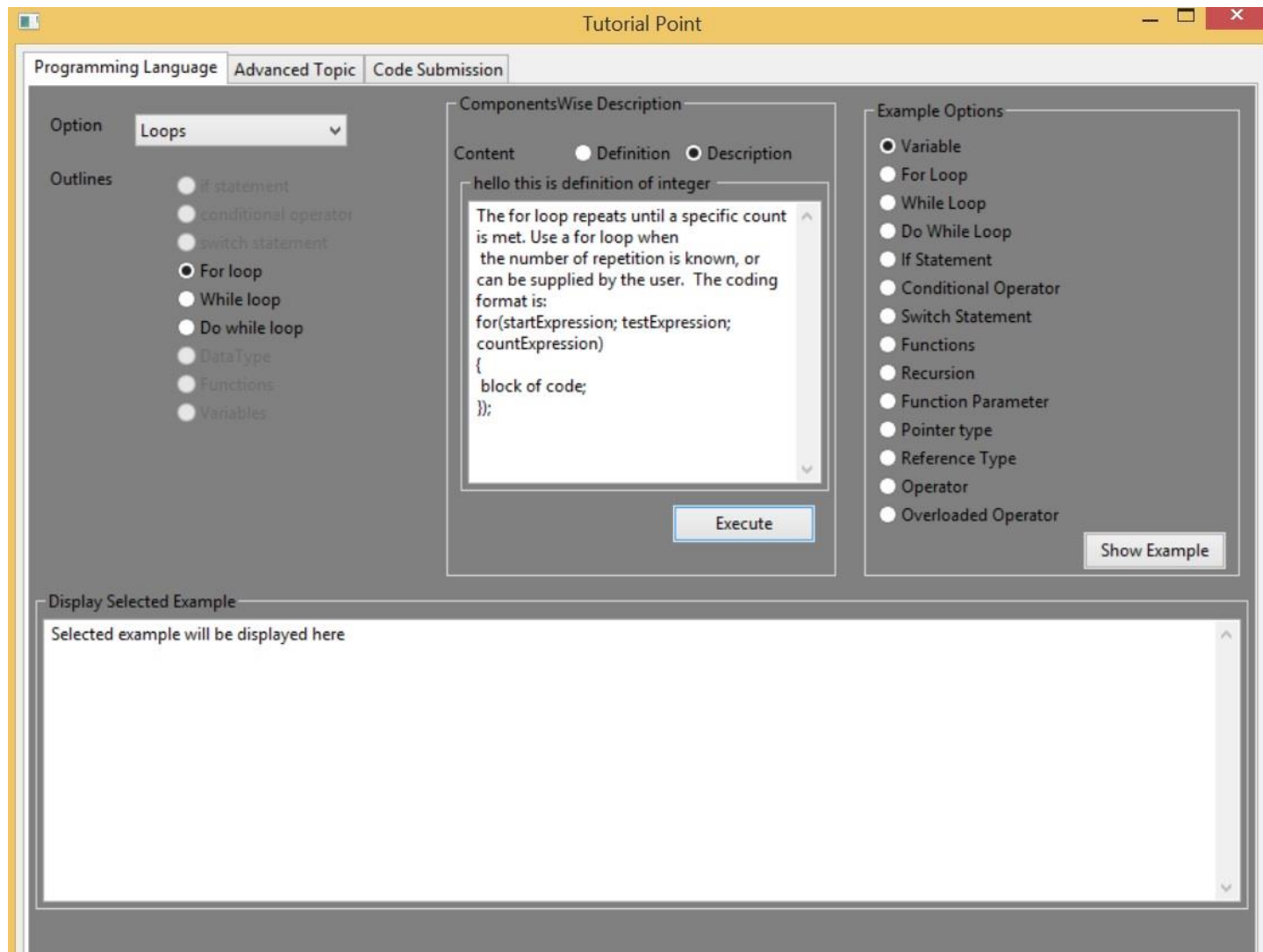


Fig 6:

User can also select an example option to understand the different aspect of programming language domain. Following image shows the presentation of For Loop example,

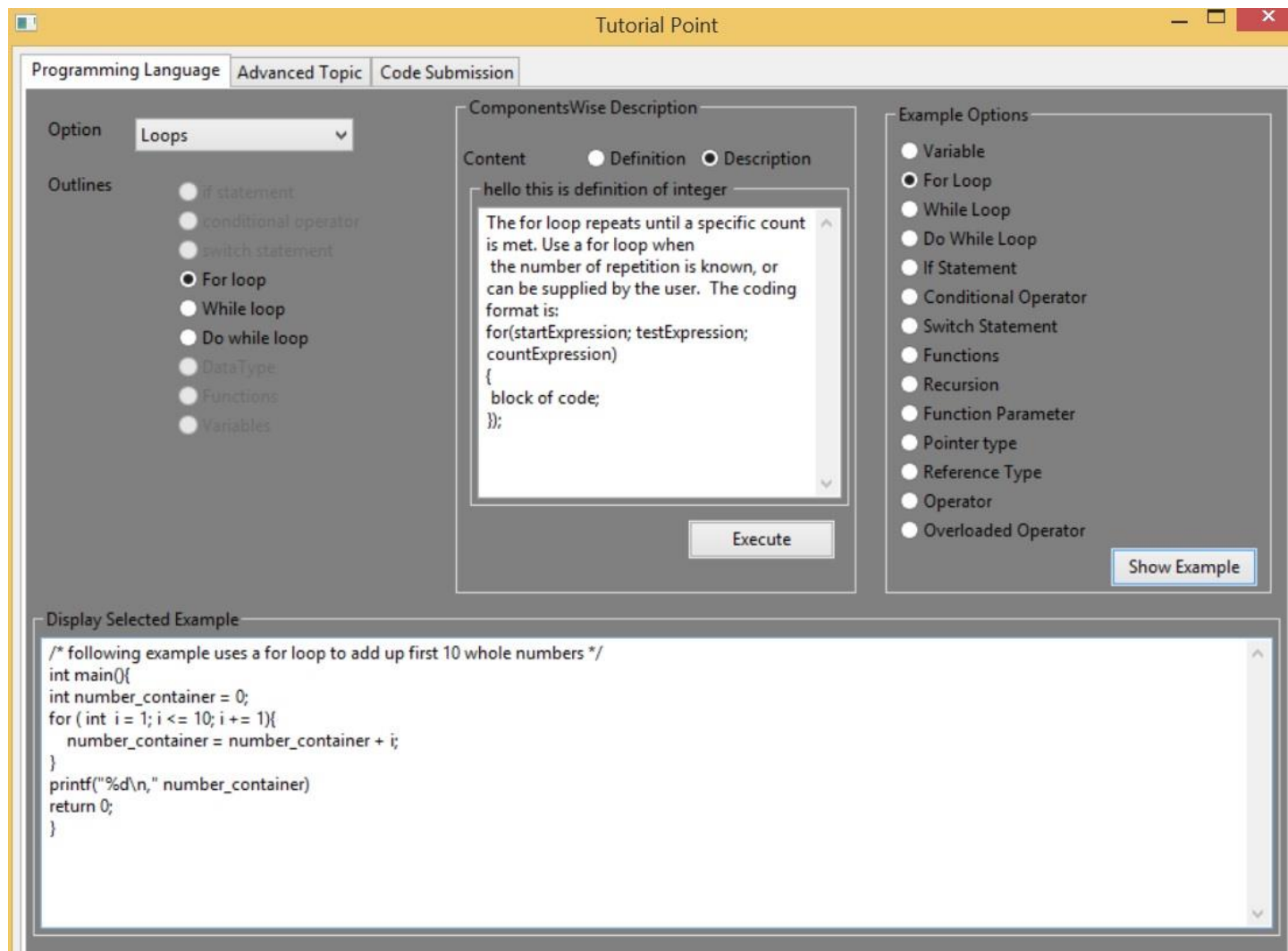


Fig 7:

Introduction with IDE and Writing Solution:

After learning the different aspects of programming language the next step of the tutorial is to teach the students about the Integrated Development Environment by using which they are going to write the programming code for different mathematical and IOI problems. There should be an option in this application where user will find details of how to install IDE and how to open a windows console project in it. However I did not develop and attach that option in the application therefore I am going to describe how it can be done using the following description.

For visual studio installation just need to do the following steps,

1. Download the visual studio from this Microsoft link <http://www.visualstudio.com/en-us/products/free-developer-offers-vs.aspx>

2. Then click on the installation button and perform the custom installation by just clicking the next button
3. It may take more than hours to complete the installation process

Now students are ready to open a windows console application using the visual studio. To do that students have to perform the following steps,

1. User has to double click the short cut of visual studio to open it. He will get the following window,

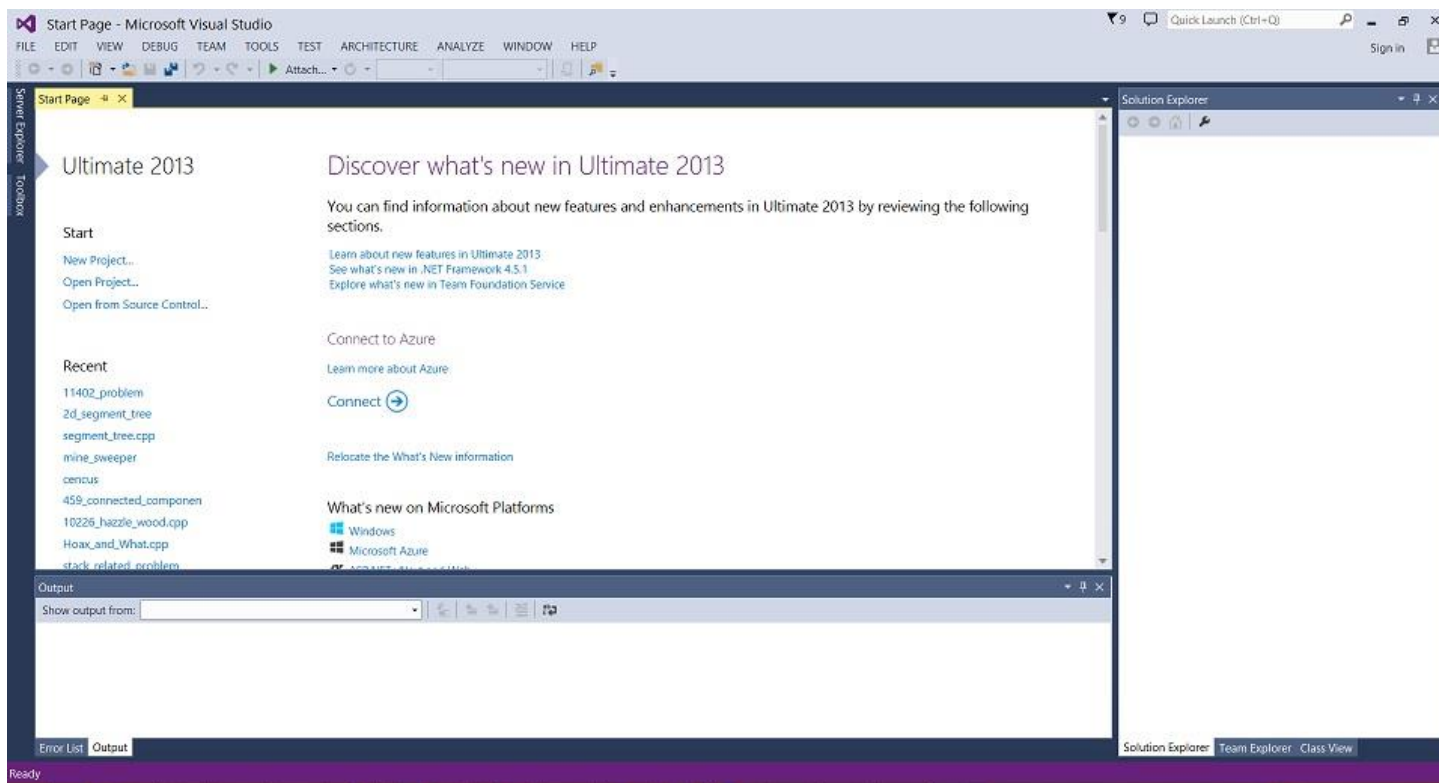


Fig 8:

2. Now click the File option at the menu bar and Open New

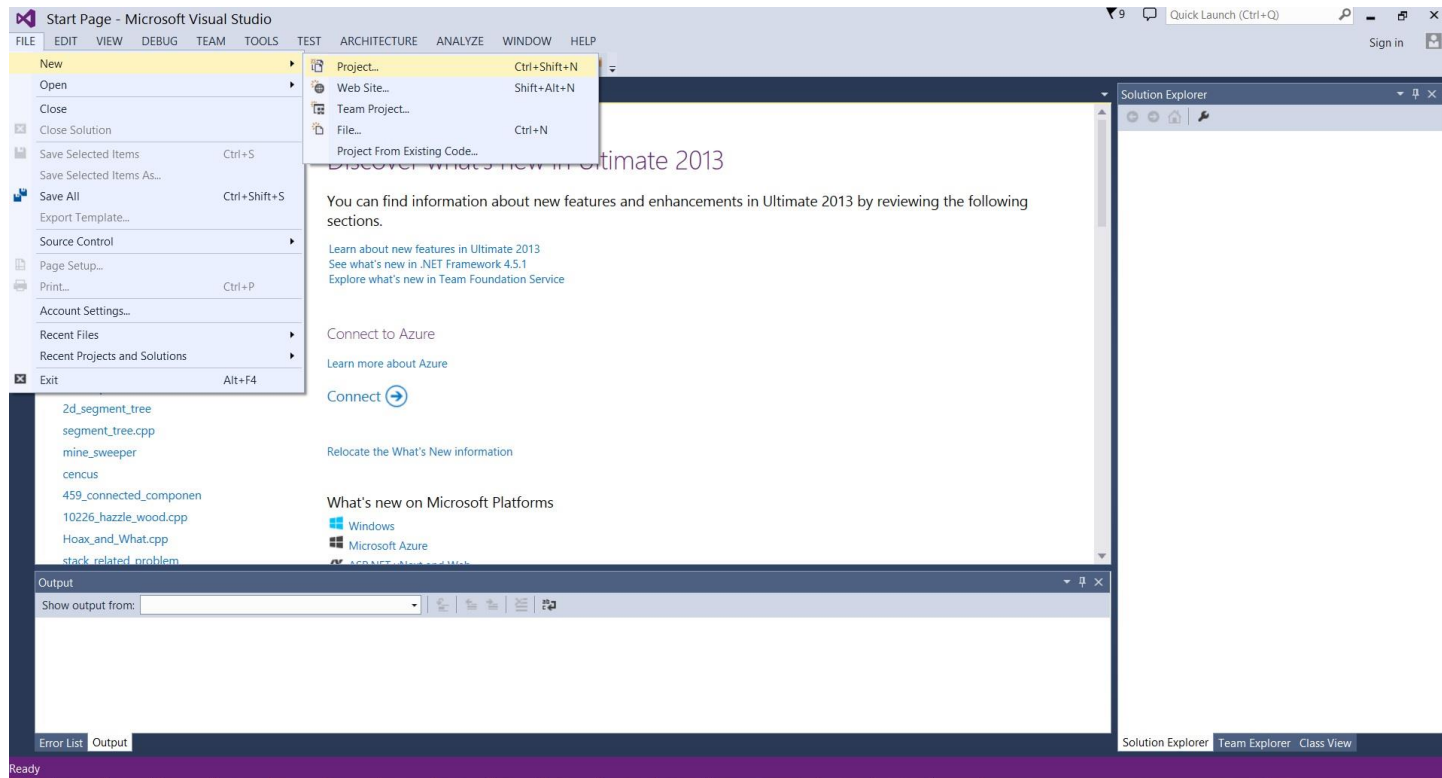


Fig 9:

3. When user opens a new project visual studio will give a template where all the installed language will be visible. User has to select his preferable language. For now it is going to be visual C++.

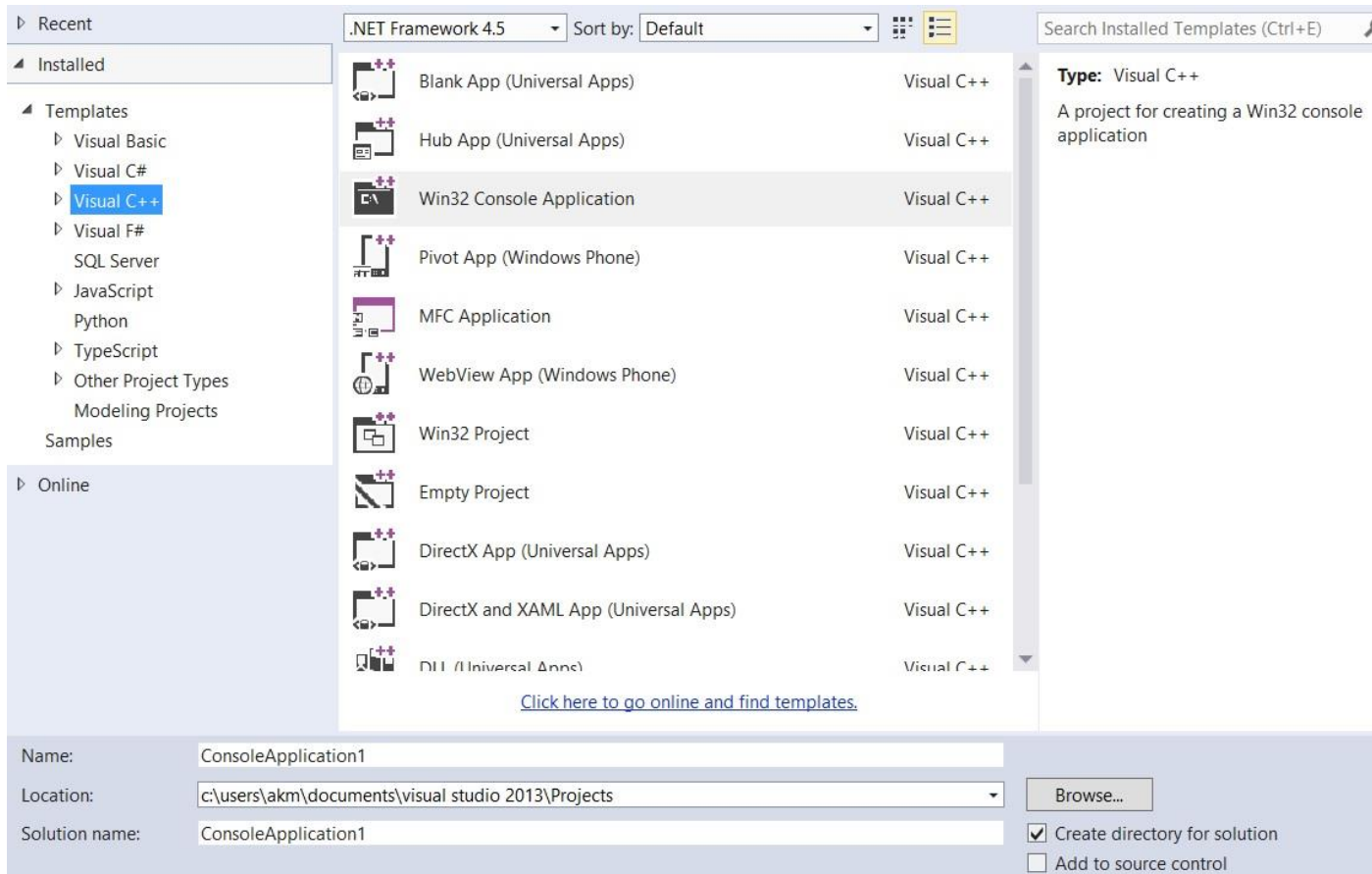


Fig 10:

4. After selection of a language; user has to give a valid name of the project and has to select win 32 console application option then press ok button

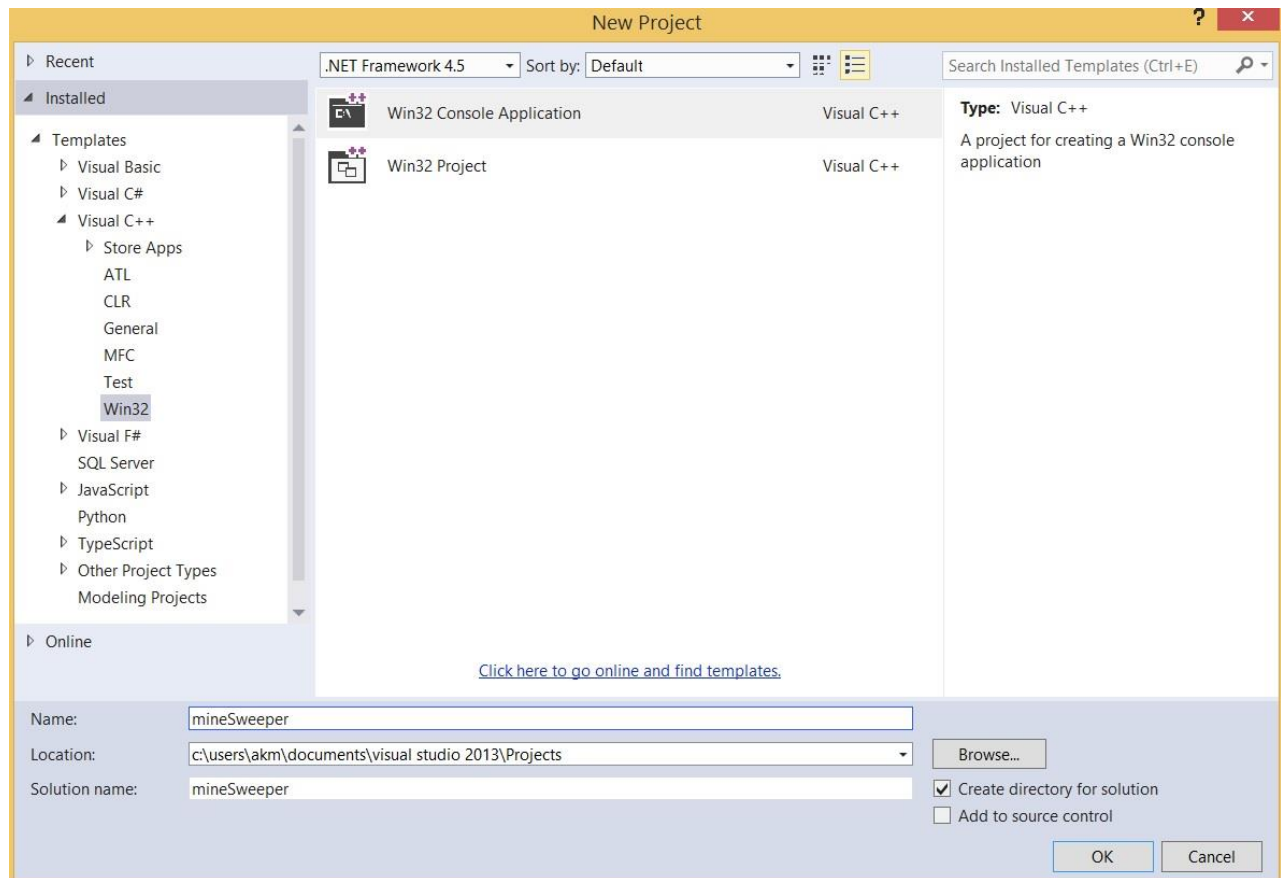


Fig 11:

5. When user presses ok it will open the win 32 application wizard. Then user has to press the next button and configure the project as follows. He has to select the empty project option. Deselect the security development life cycle and select the console application option. As it is shown in following image

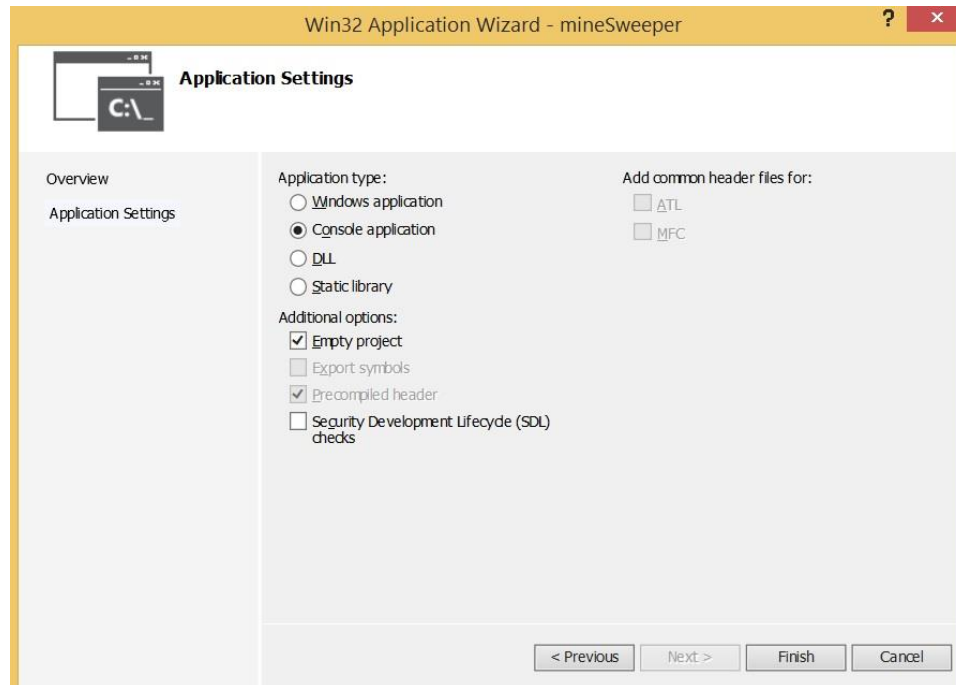


Fig 12:

6. Now pressing the finish button will open the project and the solution explorer will look like this

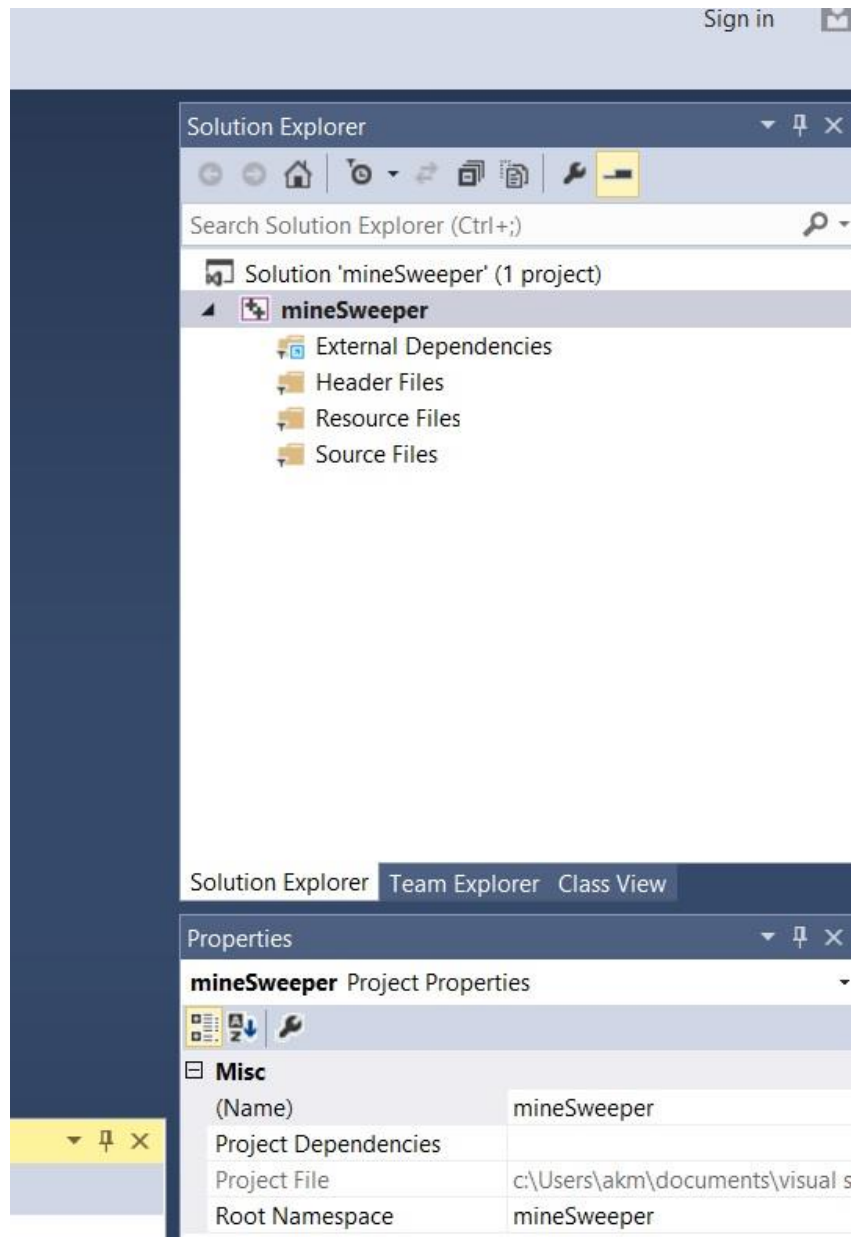


Fig 13:

- Now right click on the Source Files directory and choose add new item to add a new C++ source file and named it main.cpp

I have written a solution for MineSweeper problem. This problem has been taken from ACM UVA site. This site contains thousands of problems can be solved by using programming language. Some of them are IOI level. MineSweeper is one of the easy level problems. Students can expect to face problem like this in International Olympiad in Informatics. The problem description is given below,

Have you ever played Minesweeper? It's a cute little game which comes within a certain Operating System which name we can't really remember. Well, the goal of the game is to find where are all the mines within a $M \times N$ field. To help you, the game shows a number in a square which tells you how many mines there are adjacent to that square. For instance, suppose the following 4×4 field with 2 mines (which are represented by an '*' character):

```
*...
....
.*..
....
```

If we would represent the same field placing the hint numbers described above, we would end up with:

```
*100
2210
1*10
1110
```

As you may have already noticed, each square may have at most 8 adjacent squares.

Input

The input will consist of an arbitrary number of fields. The first line of each field contains two integers n and m ($0 < n, m \leq 100$) which stands for the number of lines and columns of the field respectively. The next n lines contains exactly m characters and represent the field.

Each safe square is represented by an '.' character (without the quotes) and each mine square is represented by an '*' character (also without the quotes). The first field line where $n = m = 0$ represents the end of input and should not be processed.

Output

For each field, you must print the following message in a line alone:

Field # x :

Where x stands for the number of the field (starting from 1). The next n lines should contain the field with the '.' characters replaced by the number of adjacent mines to that square. There must be an empty line between field outputs.

Fig 16:

Solution for above problem written in C++ using visual studio 2013 is,

```
#include<stdio.h>
#include<iostream>
#include<string>
#include<vector>
using namespace std;
int find_mines(vector<string> source,int x,int y,int sizex,int sizey){
    if (x < 0 || y < 0)
        return 0;
    if (x == sizex || y == sizey)
        return 0;
```

```

        if (source[x][y] == '*')
            return 1;
        if (source[x][y] == '.')
            return 0;
        return 0;
    }
    int main(){

        vector<string> str_container;
        vector<string> game_board;
        vector<int> just_test;
        just_test.resize(1000);
        printf("%d\n",just_test.capacity());
        printf("%d\n",just_test.size());
        printf("%d\n",just_test.at(1));
        int cases;
        string str;
        int count = 0;
        int sizex,sizey;
        while (true){
            getline(cin, str);
            if (str.empty())
                continue;
            sscanf(str.c_str(), "%d%d", &sizex,&sizey);
            if (sizex == 0 || sizey == 0)
                break;
            char result[101][101];
            for (int i = 0; i < sizex; i++){
                getline(cin, str);
                str_container.push_back(str);
            }

            int mine_touched = 0;
            for (int i = 0; i < sizex; i++){

                for (int j = 0; j < sizey; j++){
                    int count = 0;
                    if (str_container[i][j] == '.'){
                        count += find_mines(str_container, i - 1, j - 1, sizex,sizey);
                        count += find_mines(str_container, i, j - 1, sizex, sizey);
                        count += find_mines(str_container, i, j + 1, sizex, sizey);
                        count += find_mines(str_container, i - 1, j, sizex, sizey);
                        count += find_mines(str_container, i + 1, j, sizex, sizey);
                        count += find_mines(str_container, i + 1, j + 1, sizex, sizey);
                        count += find_mines(str_container, i + 1, j - 1, sizex, sizey);
                        count += find_mines(str_container, i - 1, j + 1, sizex, sizey);
                        result[i][j] = (char)(count + 48);
                    }
                    else{
                        result[i][j] = '*';
                    }
                }
            }
        }
    }
}

```

```

        }
        if (mine_touched != 0 && str_container[i][j] == '*')
            result[i][j] = '*';
    }
}
if (count != 0)
    printf("\n");
printf("Field #%%d:\n", count + 1);
for (int i = 0; i < sizex; i++){
    for (int j = 0; j < sizey; j++)
        cout << result[i][j];
    cout << endl;
}
count++;
str_container.clear();
}
return 0;
}

```


Solution Submission to Online Judge Using Application and Online Site:

In order to check the solution whether the written code compile, run and generate the correct output students need to submit the solution to online judge. They can do that by using the developed application in the following way,

The screenshot shows a window titled "Tutorial Point" with three tabs: "Programming Language", "Advanced Topic", and "Code Submission". The "Code Submission" tab is active. It contains a "File Upload and Clarification Query" section with a "Browse and Select File" button, a text field showing a file path, and a "Browse" button. Below this is a "Clarification Query" text field. To the right is a "Select Options" section with a "ProblemID" text field containing "10189" and two radio button options: "C++ 4.8.2 - GNU C++ Compiler with options: -l" (selected) and "JAVA 1.7.0 - Java Sun JDK". A "Submit" button is located below these options. At the bottom is a "List of Submission Result" section containing a table with 18 rows and 6 columns: "Submission #", "Problem Name", "Verdict", "Language", and "RunTime". The first two rows are filled with data, and the rest are empty.

	Submission #	Problem Name	Verdict	Language	RunTime
1	108923	cencus	accepted	c++	1.0 sec
2	105909	MineSweeper	accepted	c++	2.5 sec
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					

Fig 16:

Browse and select the file by clicking the browse button. Then write down problem ID will be given for each problem from select option section at same time students will also need to select the type of compiler depending on the programming language they are going to use to write solution. Finally press the submit button. Students will be able to see all submission result they made using their account.

The online judge page from ACM UVA is look like the below image. The IOI online Judge system is almost similar to this one,

Google™ Custom Search

Search

Main Menu

Home

My Account

Contact Us

ACM-ICPC Live Archive

Logout

Online Judge

Quick Submit

Migrate submissions

My Submissions

My Statistics

My uHunt with Virtual Contest Service

Browse Problems

Quick access, info and search

Problemsetters' Credits

Live Rankings

Site Statistics

Contests

Electronic Board

Additional Information

Other Links



Quick Submit

Problem ID

Language

☐ ANSI C 4.8.2 - GNU C Compiler with options: -lm -lcrypt -O2 -pipe -ansi -DONLINE_JUDGE

☐ JAVA 1.7.0 - Java Sun JDK

☐ C++ 4.8.2 - GNU C++ Compiler with options: -lm -lcrypt -O2 -pipe -DONLINE_JUDGE

☐ PASCAL 2.6.2 - Free Pascal Compiler

☐ C++11 4.8.2 - GNU C++ Compiler with options: -lm -lcrypt -O2 -std=c++11 -pipe -DONLINE_JUDGE

Paste your code...

...or upload it

Browse...

No file selected.

Submit

Reset form

Fig 18:

The verdict for minesweeper problem that I have submitted is,

Main Menu

[Home](#)
[My Account](#)
[Contact Us](#)
[ACM-ICPC Live Archive](#)
[Logout](#)

Online Judge

[Quick Submit](#)
[Migrate submissions](#)
[My Submissions](#)
[My Statistics](#)
[My uHunt with Virtual Contest Service](#)
[Browse Problems](#)
[Quick access, info and search](#)
[Problemsetters' Credits](#)
[Live Rankings](#)
[Site Statistics](#)
[Contests](#)
[Electronic Board](#)
[Additional Information](#)
[Other Links](#)

My Submissions

#	Problem	Verdict	Language	Run Time	Submission Date
15035636	11402 Ahoy, Pirates!	Time limit exceeded	C++	5.000	2015-02-25 00:38:57
15030806	11402 Ahoy, Pirates!	Time limit exceeded	C++	5.000	2015-02-24 03:30:14
15030805	11402 Ahoy, Pirates!	Compilation error	C++	0.000	2015-02-24 03:30:08
15030767	11402 Ahoy, Pirates!	Time limit exceeded	C++	5.000	2015-02-24 03:16:21
15030720	11402 Ahoy, Pirates!	Runtime error	C++	0.000	2015-02-24 03:05:14
15016091	10189 Minesweeper	Accepted	C++	0.555	2015-02-21 08:04:24
15015943	10279 Mine Sweeper	Accepted	C++	0.042	2015-02-21 07:25:06
15015922	10279 Mine Sweeper	Accepted	C++	0.025	2015-02-21 07:21:06
15015886	10279 Mine Sweeper	Wrong answer	C++	0.052	2015-02-21 07:11:26
15015854	10279 Mine Sweeper	Wrong answer	C++	0.032	2015-02-21 07:04:28
15015548	11297 Census	Accepted	C++	1.008	2015-02-21 05:59:34
15010187	459 Graph Connectivity	Accepted	C++	0.035	2015-02-20 04:00:38
15009686	459 Graph Connectivity	Wrong answer	C++	0.029	2015-02-20 00:45:13
15009623	459 Graph Connectivity	Wrong answer	C++	0.029	2015-02-20 00:23:08
15009511	459 Graph Connectivity	Wrong answer	C++	0.032	2015-02-19 23:37:44
15009492	459 Graph Connectivity	Wrong answer	C++	0.036	2015-02-19 23:32:17
15009441	459 Graph Connectivity	Wrong answer	C++	0.035	2015-02-19 23:18:49
15009418	459 Graph Connectivity	Wrong answer	C++	0.035	2015-02-19 23:10:58
15007777	11235 Frequent values	Accepted	C++	0.942	2015-02-19 17:16:09
15007742	11235 Frequent values	Wrong answer	C++	0.928	2015-02-19 17:08:33

Fig 20:

The accepted verdict is marked with black color.

Learning Outcome:

There will be several learning outcome for students if they use this 'tutorial point' application. They are

- Students will learn about structural programming
- They will learn about different aspect of programming languages
- They will be able to write programs and solve critical thinking problems
- They will learn about Integrated Development Environment
- They will learn about visual studio and how to use it to write win32 console program
- They will be able to prepare for participating in International Olympiad in Informatics
- It will help students to develop their critical thinking level and prepare them to solve more challenging problems

Learning Outcomes in terms of Software Engineering:

Things the students are going to learn about software engineering are graphical user interface, code writing style, debugging process, testing process and authentication process. They will learn about properly designing a graphical user interface to simulate a real life problem. Students know that they are going to use this application to simulate the environment they are going to experience in the International Olympiad competition. So all the design of GUI of this application is kind of mimic the IOI environmental setup. So students are going to learn about implementation of simulation environment using software engineering design process.

They will also learn about how to put different options on the GUI template to ease the user interaction with the application. When write solution for different problems using programming language they learn about the proper

way to write down code that is coding style. They will learn about debugging process when there is something wrong about the solution or in programming code.